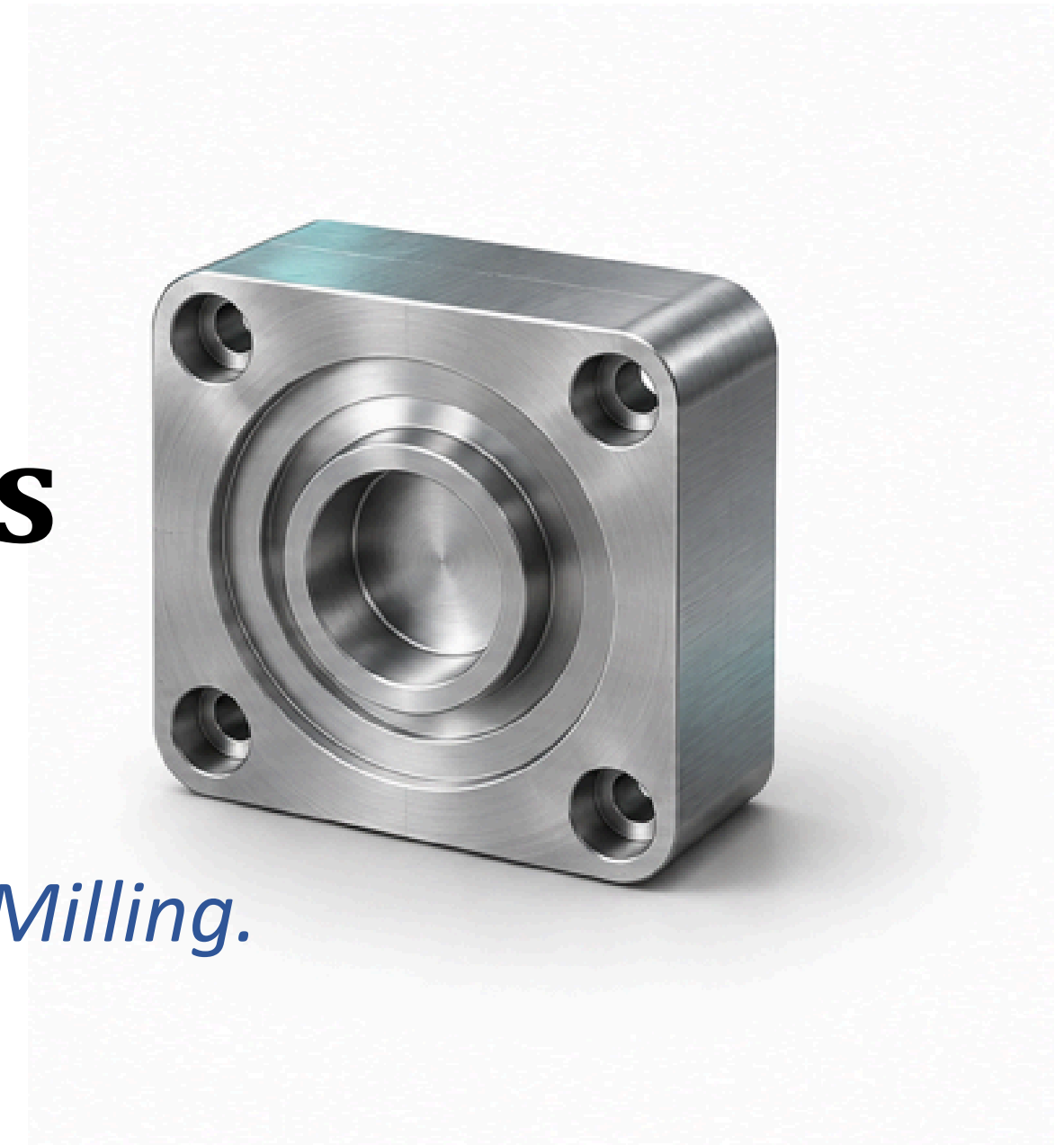


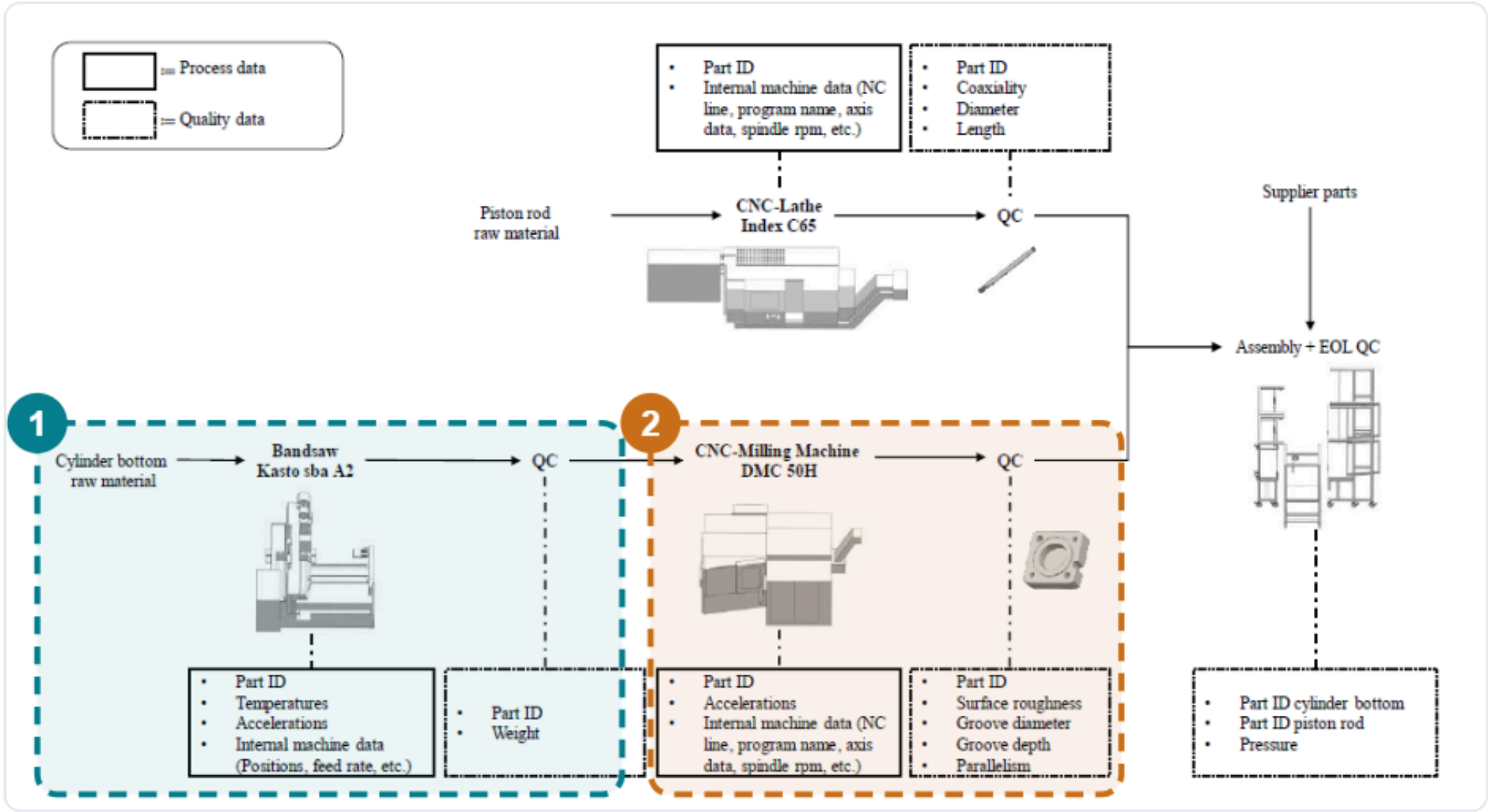
Saw-Stage Early Warning for Hidden Downstream Defects

*A workpiece can pass the Saw weight check and still fail at Milling.
We catch it before Milling starts.*



The defect is made at the Saw, but only found after Milling

Weight is the only Saw-stage gate — and a short blank can still weigh correctly.



What the data shows

- 99** Saw misalignments recorded
- 100%** passed the Saw weight check
- 81** later failed Milling QC

Why it matters

When the detected parts move forward milling time, tooling and energy are spent and no signal is given before milling stage to identify possible detected ones.

1 Made here

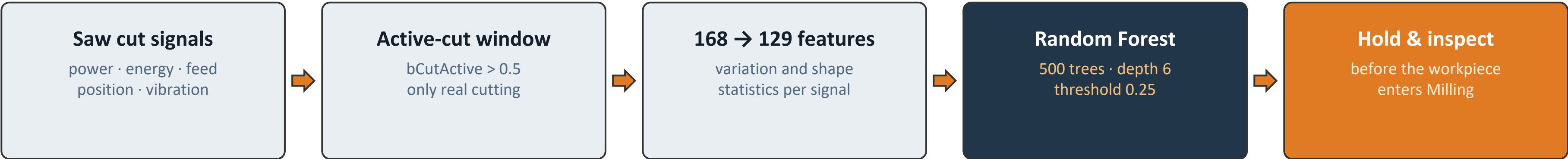
The Saw fault passes the weight check.

2 Found here

Only after Milling effort is spent.

A risk score at the moment the Saw finishes cutting

Only Saw process signals go in. Downstream quality defines the historical target — never the input.



Target: a hidden Saw-originated quality escape

Saw misalignment introduced

AND

Passed Saw weight QC

AND

Failed Milling QC

Everything else with a known outcome is a negative — including products that fail Milling for other reasons.

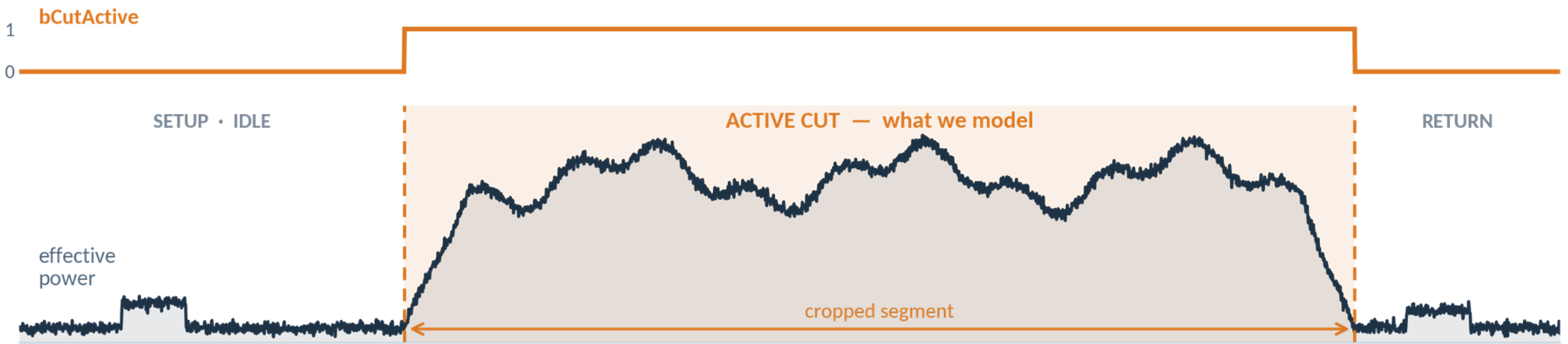
Scope we claim

An inspection recommendation, not an automatic scrap or release decision. Existing QC stays in place. This is a retrospective prototype, not a deployed system.

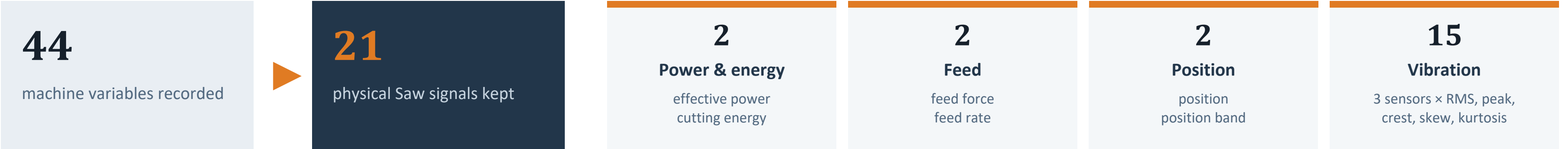
The existing Saw weight QC catches 0 of these 30 target defects by construction — every one of them already passed that inspection.

From raw Saw telemetry to the signals that describe cutting

984 prepared recordings, 44 machine variables, complete sequences — only part of that physically represents the cut.



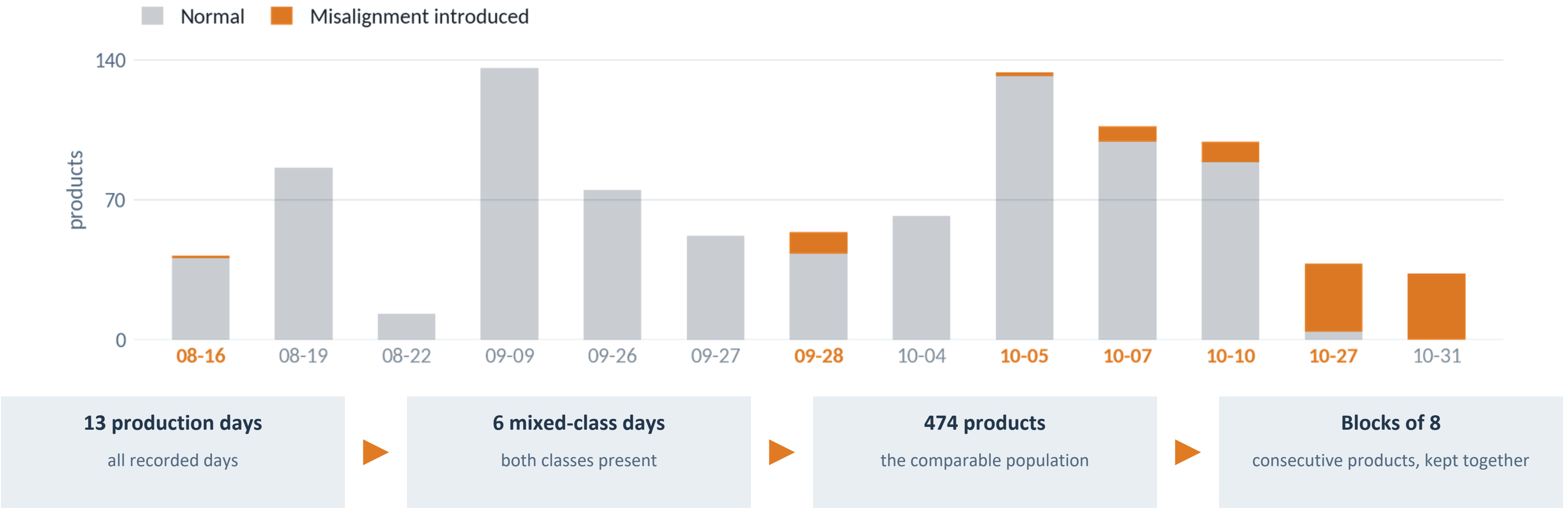
Which variables we kept as model inputs



Excluded from the predictors: IDs · timestamps · QC outcomes · fault labels · machine-state counters

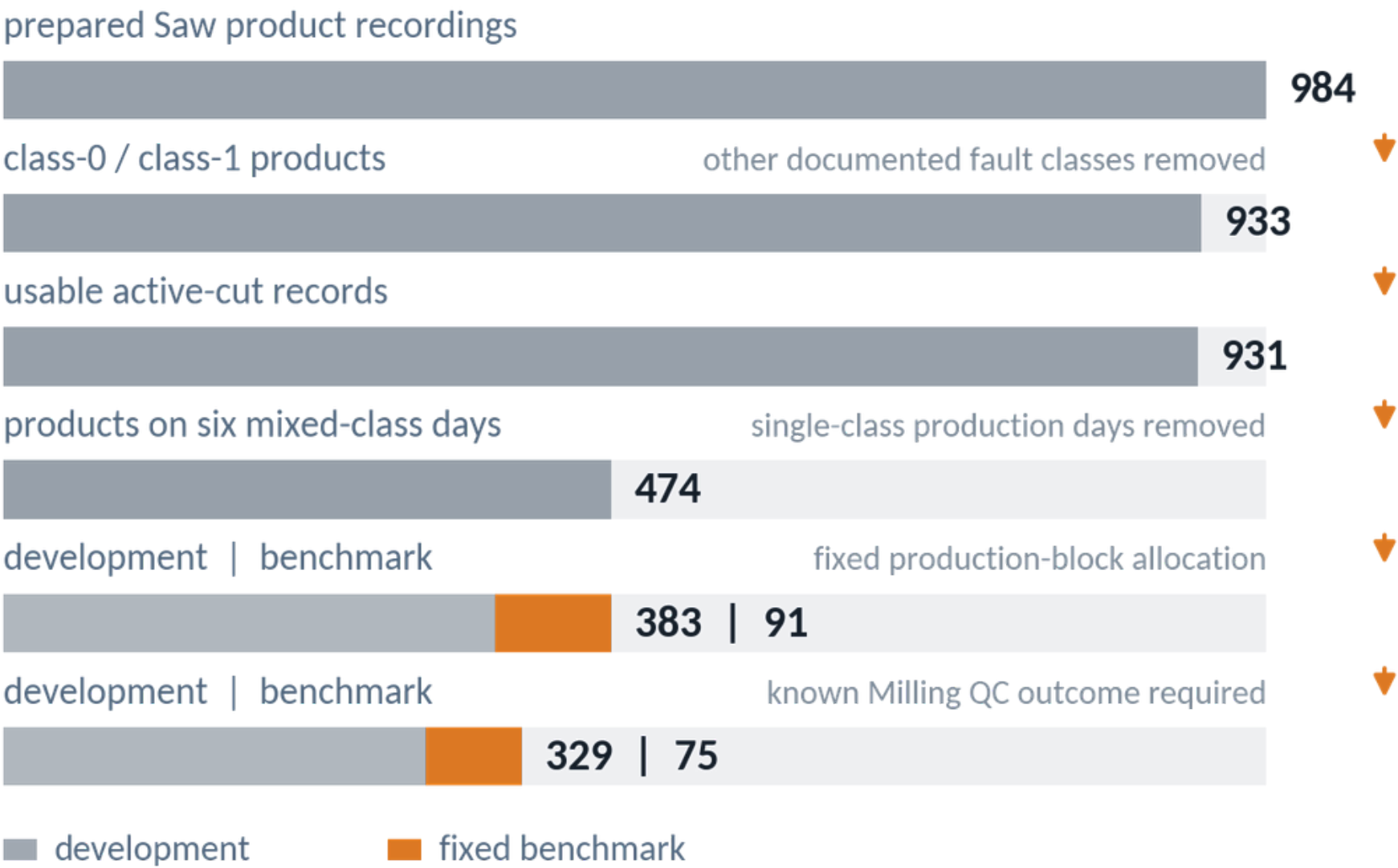
Production time revealed a proxy-learning risk

The two classes are not evenly distributed across production days — so process signals could encode when a product was made.

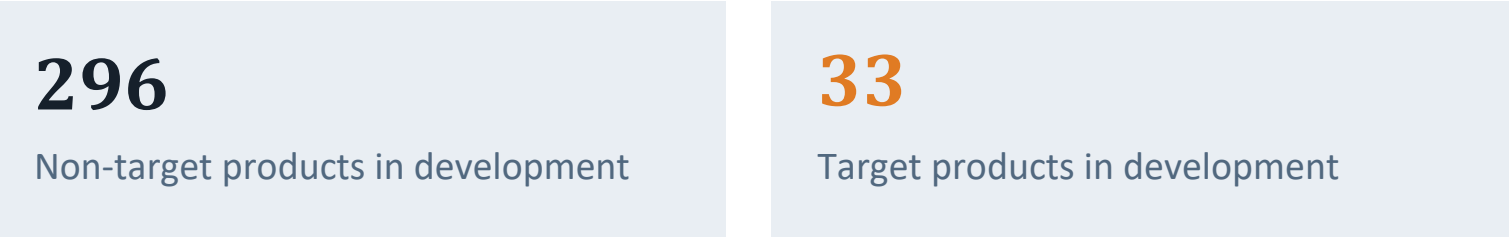


Reduces production-condition confounding — it does not eliminate it.

Building the final modelling population and target



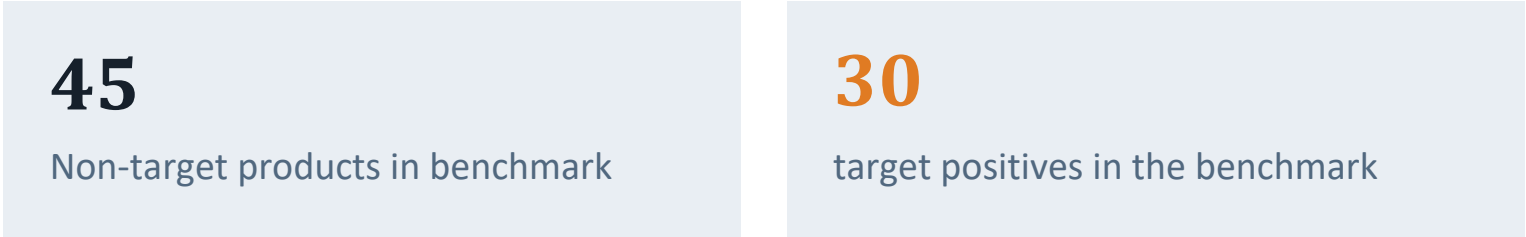
Development set



The positive target



Fixed benchmark



Two ways to represent the same cutting process

Same population, same target, same fixed split — we varied only how the active cut is represented.



Random Forest: how it was built, and how it scored

Folds, preprocessing and the operating point all happen inside the 329 development products; the benchmark is scored once.

Built inside the 329 development products

- 1

5-fold StratifiedGroupKFold

groups = production blocks · a block never straddles a fold
- 2

Stratified because targets are 1 in 10

each fold holds six or seven of the 33 targets — ten folds would leave three
- 3

Preprocessing learned inside each training fold only

drop > 20% missing · drop near-constant · impute fold median · drop $|r| > 0.95$
- 4

Out-of-fold probability for every product

thresholds 0.05 → 0.80 searched — 0.25 gives most capture at FAR ≤ 5%
- 5

Refit on all 329 after the threshold was fixed

168 → 129 features · 500 trees, depth 6, min leaf 2, balanced weights

Scored once on the 75 fixed benchmark products

	Flagged	Passed
Hidden defect · 30	20 67% of row	10 33% of row
Non-target · 45	1 2% of row	44 98% of row
	66.7% Capture / recall	95.2% Precision
	2.22% False-alarm rate	0.914 ROC-AUC
		0.784 F1 score
		0.897 Avg precision

0 / 30

found by the existing Saw weight QC — all 30 had already passed it.

The raw time-series challenger

Same population, same target, same fixed split — only the representation of the cut changes.

How the sequence was prepared

- 1

12 raw physical channels — power, energy, feed, position, vibration
- 2

Active-cut segment only, the same window the forest uses
- 3

Resampled to 128 time points, normalised per channel
- 4

Tensor $404 \times 12 \times 128$ — same split and target as the forest

On the fixed benchmark

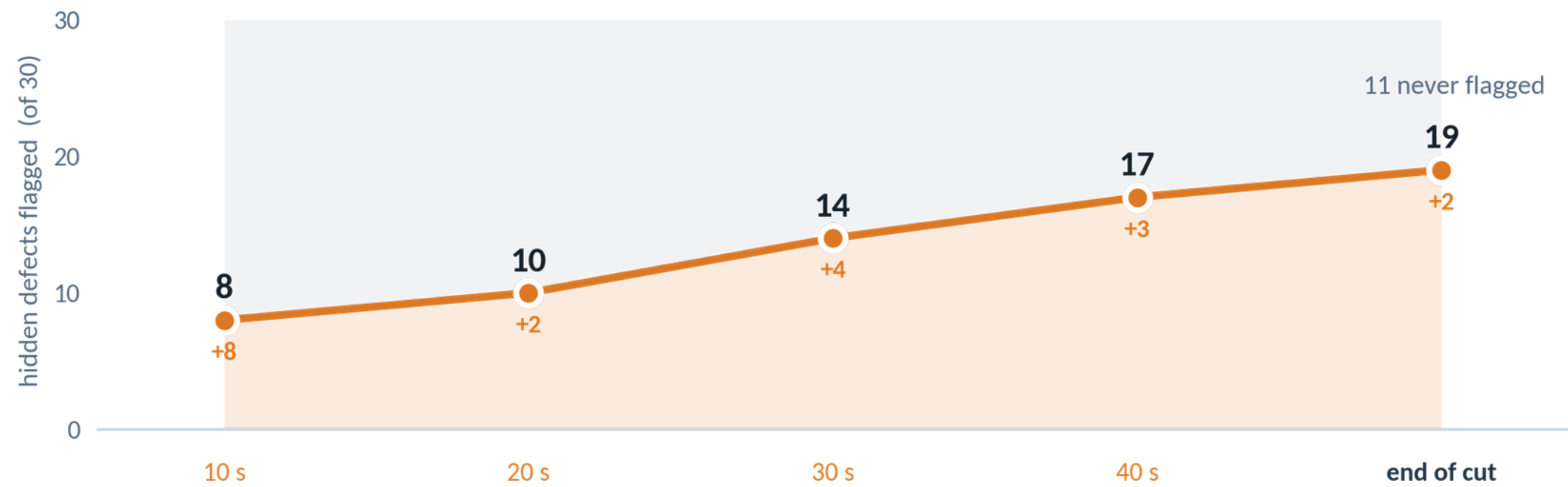
	RF	TCN
75 products		
Hidden defects caught	20 / 30	14 / 30
False alarms	1 / 45	0 / 45
Capture	66.7%	46.7%
F1	0.784	0.636
ROC-AUC	0.914	0.762
Average precision	0.897	0.812

TCN architecture

- Input convolution 12 → 32 channels, kernel 5
- 4 residual blocks, 32 channels, dilations 1 · 2 · 4 · 8
- GroupNorm + SiLU + dropout 0.15 · AdamW · class-weighted loss

Detecting before the cut finishes

Fixed-time prefixes at 10, 20, 30 and 40 seconds, using only the sensor data available at each point.



17 / 30

flagged while the Saw was still cutting

20 s

median first warning after the cut began

55.77 s

overall mean active-cut duration

2 / 45

non-target products falsely flagged

Where this sits in the line, and what it is worth

An inspection decision at a moment that already exists, using sensors that are already installed.



Reduces QC Blind Spots

Current weight-based Saw QC misses misalignment defects, while the model detects **66.7%** of them



Stops waste early

Detects defective parts before Milling, saving **machine time, energy, and labor**.



Low false alarms

: Only **2.22% false-alarm rate (1/45)**, minimizing unnecessary inspection or rework.



No new hardware

Uses existing **saw-machine sensor data**, so no additional sensors or major hardware investment are needed



Real-time warning

Can detect problems around **20 seconds into the cut**, allowing in-process intervention.



Works with existing QC

: Acts as an **early-warning layer**, complementing rather than replacing current QC.

What this evidence does not yet support

We would rather state the boundary ourselves than have it found for us.

Same production days sit in both splits

Development and benchmark draw from the same six days, so new-day, new-machine and new-factory transfer is unproven.

It also misses some real defects

Out of 30 actual hidden defects 20 was detected and 10 missed. So recall was 66.7%

The data comes from only six production days

A factory can change from day to day because of different raw materials, different machine condition, different environmental conditions, production settings

New machine / new factory is untested

The model was developed from this particular production environment. It cannot be claim yet that the model works universally across machines or factories.

Limited Material Generalization

The model was trained and tested on one material type; performance on other materials has not been validated.

23

A retrospective, end-of-Saw inspection-support prototype that uses active-cut Saw process signals to identify hidden Saw-originated quality escapes before products proceed to Milling.

20 / 30

hidden defects detected
on the fixed benchmark

1 / 45

non-target products
falsely flagged

0 / 30

caught by the existing
weight inspection

Not yet

validated on new periods,
conditions or machines

24 *Independent validation on new production periods, different operating conditions, and ideally additional machines or factories would still be required before production deployment.*

Sources and disclosures

CiP-DMD dataset paper

Jourdan et al., Procedia CIRP 126 (2024), 423–428
<https://doi.org/10.1016/j.procir.2024.08.390>

Related research

SSMSPC: <https://doi.org/10.1007/s10845-023-02156-7>
Manufacturing reliability: <https://arxiv.org/abs/2112.06986>

Data and project evidence

https://github.com/HannesJoern/interq_data_acquisition_public
Author manuscript: <https://doi.org/10.5281/zenodo.8420132>
Supplied notebook, technical report and OctWave final-round booklet